

Quiz - #1 - Weighted & Delayed Steps and Ramps

Write expressions for each signal in terms of weighted and delayed unit steps u_s and ramps u_r .

(a) Piecewise linear signal:

Flat 0 for $t \in [0, 1)$; ramp +1 at $t=1$; slope changes $-\frac{1}{2}$ at $t=2, 3, 4, 5$; +1 at $t=6$; +2 at $t=7$; -2 at $t=8, 9$.

$$u = u_r(t-1) - \frac{1}{2}u_r(t-2) - \frac{1}{2}u_r(t-3) - \frac{1}{2}u_r(t-4) - \frac{1}{2}u_r(t-5) + u_r(t-6) + 2u_r(t-7) - 2u_r(t-8) - 2u_r(t-9)$$

(b) Signal with step and ramp components:

Step up at $t=0$; ramp $-\frac{1}{2}$ at $t=2$; $+\frac{1}{2}$ at $t=4, 5$; step down at $t=4$; $+\frac{3}{2}$ at $t=7$; -2 at $t=8$; step down at $t=9$.

$$u = u_s(t) - \frac{1}{2}u_r(t-2) + \frac{1}{2}u_r(t-4) - u_s(t-4) + \frac{1}{2}u_r(t-5) + \frac{3}{2}u_r(t-7) - 2u_r(t-8) - 2u_s(t-9)$$

Quiz - #2 - 2nd Order System Damping Ratio Ranges

$\ddot{y} + 2\zeta\omega_n\dot{y} + \omega_n^2y = u(t)$. Identify ζ range for each step response plot.

Eigenvalues $\lambda = -\zeta\omega_n \pm \omega_n\sqrt{\zeta^2 - 1}$; $\text{Re}(\lambda) = -\zeta\omega_n$ governs decay/growth.

Plot	Behavior	ζ range
1	Monotone, no overshoot	$\zeta \geq 1$ (overdamped)
2	Overshoot, decaying osc.	$0 < \zeta < 1$ (underdamped)
3	Sustained oscillation	$\zeta = 0$ (undamped)
4	Growing oscillation	$-1 < \zeta < 0$ (neg. damped)

Quiz - #3 - 2nd Order ODE: Homogeneous + Particular + Laplace

$\ddot{y} + 2\dot{y} + 2y = u(t)$

(a-i) $u(t) = 1$, $y(0) = 1$, $\dot{y}(0) = 0$

Homogeneous: $y_h = ce^{\lambda t}$, CE $\lambda^2 + 2\lambda + 2 = 0$:

$$\lambda = \frac{-2 \pm \sqrt{4-8}}{2} = -1 \pm j \quad (\alpha = -1, \omega = 1)$$

$$y_h = e^{-t}(C_1 \cos t + C_2 \sin t)$$

Particular: constant $\Rightarrow y_p = K$; $2K = 1 \Rightarrow y_p = \frac{1}{2}$

Full: $y = e^{-t}(C_1 \cos t + C_2 \sin t) + \frac{1}{2}$

ICs: $y(0)=1$: $C_1 + \frac{1}{2} = 1 \Rightarrow C_1 = \frac{1}{2}$

$\dot{y}(0)=0$: $-C_1 + C_2 = 0 \Rightarrow C_2 = \frac{1}{2}$

$$y(t) = e^{-t}\left(\frac{1}{2} \cos t + \frac{1}{2} \sin t\right) + \frac{1}{2}$$

(a-ii) $u(t) = 11 \cos 2t$, $y(0) = 0$, $\dot{y}(0) = 0$

Homogeneous: same $y_h = e^{-t}(C_1 \cos t + C_2 \sin t)$

Particular: $y_p = K_1 \cos 2t + K_2 \sin 2t$

$$\dot{y}_p = -2K_1 \sin 2t + 2K_2 \cos 2t$$

$$\ddot{y}_p = -4K_1 \cos 2t - 4K_2 \sin 2t$$

Substitute, collect:

$$\cos 2t : -4K_1 + 4K_2 + 2K_1 = 11 \Rightarrow -2K_1 + 4K_2 = 11$$

$$\sin 2t : -4K_2 - 4K_1 + 2K_2 = 0 \Rightarrow -4K_1 - 2K_2 = 0$$

$$K_2 = -2K_1. \quad -2K_1 - 8K_1 = 11 \Rightarrow K_1 = -1.1, \quad K_2 = 2.2$$

Full: $y = e^{-t}(C_1 \cos t + C_2 \sin t) - 1.1 \cos 2t + 2.2 \sin 2t$

ICs: $y(0)=0$: $C_1 - 1.1 = 0 \Rightarrow C_1 = 1.1$

$\dot{y}(0)=0$: $-C_1 + C_2 + 4.4 = 0 \Rightarrow C_2 = -3.3$

$$y(t) = e^{-t}(1.1 \cos t - 3.3 \sin t) - 1.1 \cos 2t + 2.2 \sin 2t$$

(b) Stability:

$$\lambda = -1 \pm j \Rightarrow \text{Re}(\lambda) = -1 < 0$$

Stable — both eigenvalues have negative real part

(c) Laplace method for (a-i): $u(t) = 1$, $y(0) = 1$, $\dot{y}(0) = 0$

$\mathcal{L}\{\text{ODE}\}$: $[s^2Y - s] + 2[sY - 1] + 2Y = \frac{1}{s}$

$$Y(s^2 + 2s + 2) = \frac{1}{s} + s + 2 \Rightarrow Y = \frac{s^2 + 2s + 1}{s(s^2 + 2s + 2)} = \frac{(s+1)^2}{s[(s+1)^2 + 1]}$$

$$\text{PFD: } Y = \frac{A}{s} + \frac{\alpha s + \beta}{s^2 + 2s + 2}. \quad A = \frac{1}{2}, \text{ match: } \alpha = \frac{1}{2}, \beta = 1.$$

$$Y = \frac{1}{2s} + \frac{1}{2} \cdot \frac{s+1}{(s+1)^2 + 1} + \frac{1}{2} \cdot \frac{1}{(s+1)^2 + 1}$$

$$y(t) = \frac{1}{2} + \frac{1}{2}e^{-t} \cos t + \frac{1}{2}e^{-t} \sin t$$

(d) Critically damped ($u = 0$):

Need repeated real roots: $(2\zeta\omega_n)^2 = 4\omega_n^2$. Currently $\omega_n^2 = 2$, so need $\omega_n^2 = 1$.

$$\ddot{y} + 2\dot{y} + y = 0 \Rightarrow \lambda = -1 \text{ (repeated)}$$

HW4 - #1 - Signal Decomposition: Steps (8.2a) and Ramps (8.2b)

Decompose signals into weighted/delayed steps and ramps.

(a) Rectangular pulse signal:

Step up $t=0$, down $t=1$, down $t=2$, up $t=3$.

$$u(t) = u_s(t) - u_s(t-1) - u_s(t-2) + u_s(t-3)$$

(b) Triangular ramp signal:

Ramp +1 at $t=1$, change -2 at $t=2$, +1 at $t=3$.

$$u(t) = u_r(t-1) - 2u_r(t-2) + u_r(t-3)$$

HW4 - #2 - Repeated Roots: Homogeneous Solution (8.9c)

$\ddot{y} + 6\dot{y} + 9y = 0$, $y(0) = 1$, $\dot{y}(0) = 0$

CE: $\lambda^2 + 6\lambda + 9 = (\lambda + 3)^2 = 0 \Rightarrow \lambda_{1,2} = -3$ (repeated)

$$y = C_1 e^{-3t} + C_2 t e^{-3t}$$

$y(0)=1$: $C_1 = 1$. $\dot{y}(0)=0$: $-3C_1 + C_2 = 0 \Rightarrow C_2 = 3$.

$$y(t) = e^{-3t} + 3te^{-3t}$$

HW4 - #3 - Euler Identity & Particular Solution (8.11)

$\ddot{y} + y = 6e^{-2t} \cos 4t$

$$y_p = e^{-2t}(K_1 \cos 4t + K_2 \sin 4t)$$

$$\dot{y}_p = e^{-2t}[(-2K_1 + 4K_2) \cos 4t + (-4K_1 - 2K_2) \sin 4t]$$

Sub into $\ddot{y} + y = 6e^{-2t} \cos 4t$, divide by e^{-2t} :

$$\cos 4t : -K_1 + 4K_2 = 6$$

$$\sin 4t : -4K_1 - K_2 = 0 \Rightarrow K_2 = -4K_1$$

$$-K_1 - 16K_1 = 6 \Rightarrow K_1 = -\frac{6}{17}, \quad K_2 = \frac{24}{17}$$

$$y_p = e^{-2t}\left(\frac{24}{17} \sin 4t - \frac{6}{17} \cos 4t\right)$$

HW4 - #4 - Sinusoidal Particular Solution (8.12)

$\ddot{y} + 9\dot{y} + 20y = 5 \sin 2t$

$$y_p = K_1 \cos 2t + K_2 \sin 2t, \quad \dot{y}_p = -2K_1 \sin 2t + 2K_2 \cos 2t, \quad \ddot{y}_p = -4K_1 \cos 2t - 4K_2 \sin 2t$$

Substitute:

$$\cos 2t : 16K_1 + 18K_2 = 0$$

$$\sin 2t : -18K_1 + 16K_2 = 5$$

$$K_1 = -\frac{9}{8}K_2. \quad \frac{162}{8}K_2 + 16K_2 = \frac{290}{8}K_2 = 5 \Rightarrow K_2 = \frac{4}{29}, \quad K_1 = -\frac{9}{29}$$

$$y_p = -\frac{9}{58} \cos 2t + \frac{4}{29} \sin 2t$$

HW4 - #5 - Stability via Eigenvalue Analysis (8.18b)

$\ddot{y} + \dot{y} - ay = 0$. Find values of a for stability.

$$\text{CE: } \lambda^2 + \lambda - a = 0 \Rightarrow \lambda_{1,2} = \frac{-1 \pm \sqrt{1+4a}}{2}$$

Case 1: $1+4a < 0$ ($a < -\frac{1}{4}$): complex roots, $\text{Re}(\lambda) = -\frac{1}{2} < 0 \rightarrow$ stable.

Case 2: $1+4a \geq 0$ ($a \geq -\frac{1}{4}$): real roots. $\lambda_1 = \frac{-1+\sqrt{1+4a}}{2} < 0 \Rightarrow \sqrt{1+4a} < 1 \Rightarrow a < 0$.

Stable iff $a < 0$; marginal: $a=0$; unstable: $a > 0$

HW5 - #1 - RC Circuit: State Equation & Time Constant (9.4)

Voltage source V_s , resistors R_1 , R_2 , capacitor C . Derive state equation, time constant, and limiting cases.



State eq. (q_c = charge on C):

$$\dot{q}_c = \frac{V_s}{R_1} - \frac{R_1 + R_2}{R_1 R_2 C} q_c$$

$$\text{Standard form: } \frac{CR_1 R_2}{R_1 + R_2} \dot{q}_c + q_c = \frac{CR_2}{R_1 + R_2} V_s$$

(b) Time constant:

$$\tau = \frac{CR_1 R_2}{R_1 + R_2}. \quad R_1 = R_2 = R = 10 \Omega, \quad C = 10^{-6} \text{ F}$$

$$\tau = CR/2 = 5 \times 10^{-6} \text{ s}$$

(c) $R_2 \rightarrow \infty$:

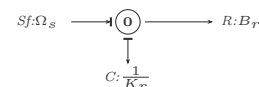
$\tau \rightarrow CR_1$. Series R_1 - C : $CR_1 \dot{q}_c + q_c = CV_s$.

(d) $R_1 \rightarrow 0$:

$\tau \rightarrow 0$. $q_c = CV_s$ (instantaneous tracking).

HW5 - #2 - Clutch-Spring: ODE & Step Response (9.6c)

Rotational spring K_r , damper B_r , input shaft Ω_s .



$$\text{ODE in torque } T_K = K_r \theta_K: \quad \frac{B_r}{K_r} \dot{T}_K + T_K = B_r \Omega_s(t)$$

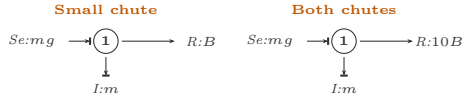
Step response ($\Omega_s = (\Omega_s)_0$, zero IC):

$$T_K(t) = B_r (\Omega_s)_0 [1 - e^{-K_r t / B_r}]$$

Steady state at $t = 4\tau = \frac{4B_r}{K_r}$.

HW5 – #3 – Parachutist: Switching Dynamics (9.12)

Mass m , drag B (small chute), $10B$ (both chutes).



ODEs: Small: $\frac{m}{B} \dot{V} + V = \frac{mg}{B}$ Both: $\frac{m}{10B} \dot{V} + V = \frac{mg}{10B}$

(b) **Velocity response:**

Phase 1 ($0 < t < 4m/B$, $\tau_1 = m/B$):

$$v(t) = \frac{mg}{B} \left[1 - e^{-Bt/m} \right] \xrightarrow{t \rightarrow 4\tau_1} V \approx \frac{0.98 mg}{B}$$

Phase 2 ($t > 4m/B$, $\hat{t} = t - 4m/B$, $\tau_2 = m/(10B)$):

$$V(\hat{t}) = \frac{mg}{10B} + \frac{9mg}{10B} e^{-10B\hat{t}/m}$$

Terminal $V \rightarrow mg/(10B)$ reached at $t \approx 4.4 m/B$.

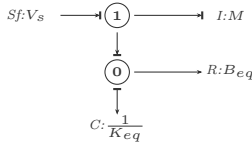
(c) **Force on skydiver $F_m = m\dot{v}$:**

Phase 1: $F_m = mg e^{-t/\tau_1}$, $\tau_1 = m/B$

Phase 2: $F_m = -9mg e^{-\hat{t}/\tau_2}$, $\tau_2 = m/(10B)$

HW5 – #4 – Spring-Mass-Damper: ω_n , η , Min Damping (9.23)

$M=320 \text{ kg}$, $K_{eq}=16000 \text{ N/m}$ ($4 \text{ springs} \times 4000$), $B_{eq}=1200 \text{ Ns/m}$.



State equations ($P=M\dot{V}_M$, X =spring deflection):

$$\begin{bmatrix} \dot{P} \\ \dot{X} \end{bmatrix} = \begin{bmatrix} -B_{eq}/M & K_{eq} \\ -1/M & 0 \end{bmatrix} \begin{bmatrix} P \\ X \end{bmatrix} + \begin{bmatrix} B_{eq} \\ 1 \end{bmatrix} V_s$$

2nd-order ODE: $\ddot{P} + \frac{B_{eq}}{M} \dot{P} + \frac{K_{eq}}{M} P = B_{eq} \dot{V}_s + K_{eq} V_s$

$$\omega_n = \sqrt{\frac{K_{eq}}{M}} = \sqrt{\frac{16000}{320}} = 5\sqrt{2} \approx 7.07 \text{ rad/s}$$

$$\eta = \frac{B_{eq}}{2\sqrt{K_{eq}M}} = \frac{1200}{2\sqrt{16000 \cdot 320}} = \frac{1200}{4525.5} \approx 0.265$$

Underdamped ($\eta < 1$)

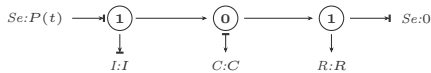
(b) **Min B per shock for no oscillation ($\eta \geq 1$):**

$$4B_{\min} = 2\sqrt{K_{eq}M} \Rightarrow B_{\min} = \sqrt{KM} = \sqrt{4000 \cdot 320}$$

$$B_{\min} = 800\sqrt{2} \text{ Ns/m}$$

HW5 – #5 – Pumping Station: Bond Graph, State Eqs, ω_n , η (9.25)

Pump $P(t)$, pipe inertance I , tank capacitance C , valve R .



State vars: Γ (pipe momentum), U_C (tank volume).

$$\begin{bmatrix} \dot{\Gamma} \\ \dot{U}_C \end{bmatrix} = \begin{bmatrix} 0 & -1/C \\ 1/I & -1/(CR) \end{bmatrix} \begin{bmatrix} \Gamma \\ U_C \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \end{bmatrix} P(t)$$

(b) **2nd-order ODE for $Q_R = U_C/(CR)$:**

$$\ddot{U}_C + \frac{1}{CR} \dot{U}_C + \frac{1}{CI} U_C = \frac{P(t)}{I}$$

$$\omega_n = \frac{1}{\sqrt{CI}}, \quad \eta = \frac{1}{2R} \sqrt{\frac{I}{C}}$$

(c) **Effect of R : $I/C = 2$.**

$$\eta = \frac{\sqrt{2}}{2R}$$

R	η	Behavior
0.5	$\sqrt{2} \approx 1.41$	Overdamped
2	$\sqrt{2}/4 \approx 0.35$	Underdamped

System changes from **overdamped** ($R=0.5$) to **underdamped** ($R=2$).

No overshoot requires $\eta \geq 1 \Rightarrow R \leq \frac{1}{2} \sqrt{I/C}$.